Varun Ravi Varma

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PERSONAL INFORMATION

Artificial Intelligence Research Student
Date of Birth: 17 August 1994
Address: Storgaten 19, H0204,
4876 Grimstad,
Norway

PERSONAL PROFILE

AI Research Scientist and Software Engineer with expertise in multi-agent reinforcement learning, interpretable decision-making, and scalable AI-driven systems. Proven track record in designing and optimizing intelligent agents using neural networks, automaton theory, and continuous control for real-world applications. Adept at bridging the gap between theoretical research and production-ready software, with extensive experience in data analytics, technical consulting, and collaborative R&D.
Languages: English (Fluent), German (A1-A2 - Learning)

EDUCATION

UNIVERSITY OF AGDER | PhD, Artificial Intelligence

Current | Grimstad, Norway

- Doctoral Researcher specializing in Multi-Agent Systems (MAS), Reinforcement Learning (RL), and model interpretability.
- Developing algorithms for path planning, resource allocation, and trajectory analysis with a focus on emergent dynamics in complex systems.

UNIVERSITY OF GRONINGEN | M.Sc. Artificial Intelligence

2019 - 2021 | Groningen, Netherlands

- Specialized in Deep Learning, Automaton Theory, and Reinforcement Learning.
- Thesis: Developed an interpretable RL agent using the Regression Tsetlin Machine to provide transparent, logic-based decision-making.

VISVESWARAYA TECHNOLOGICAL UNIVERSITY | B.E Mechanical Engineering

2012 - 2016 | Karnataka, India

- Focused on Thermodynamics, Computer-Aided Design (CAD), and mechanical component analysis.
- Developed a Peltier-effect solar cell for the final year engineering project.

EXPERIENCE

SANO CENTRE FOR COMPUTATIONAL MEDICINE | Scientific Programmer - Personal Health Data Science
June 2022 - September 2023 | Krakow, Poland

- Architected analytical pipelines for medical datasets, leveraging knowledge graphs to infer data relationships for clinical research.
- Developed and deployed research tools designed for efficient data consumption and scientific query execution.

UNIVERSITY OF GRONINGEN | Graduate Teaching Assistant
February 2020 - August 2020 | Groningen, Netherlands

- Mentored students in Deep Learning implementations and evaluated technical practical reports.
- Facilitated tutorials on the logical aspects of Multi-Agent Systems and reviewed complex course projects.

NEUDESIC GLOBAL SERVICES LLC | Consultant II - Data Analytics
July 2018 - November 2018 | Bangalore, India

- Engineered scalable solutions for large-scale data cleansing and deep learning applications.
- Designed Proof-of-Concept (PoC) ML models for global clients in the Energy and Finance sectors.
- Managed structured and unstructured data migration and analysis across cloud and on-premise infrastructures.
- Founded and led an internal upskilling group, conducting weekly sessions on statistics and machine learning.

NEUDESIC GLOBAL SERVICES LLC | Consultant I - Data Analytics
July 2017 - June 2018 | Bangalore, India

- Executed data warehousing and advanced analytics projects for diverse industry clients.
- Organized office-wide hackathons and technical workshops to drive the adoption of emerging technologies.
- Built and demonstrated various technical hobby projects and internal Proof-of-Concepts.

NEUDESIC GLOBAL SERVICES LLC | Associate Consultant
June 2016 - June 2017 | Bangalore, India

- Developed PowerShell automation scripts and maintained web application architectures for client projects.

RESEARCH PROJECTS

MASTER'S THESIS | Interpretable Reinforcement Learning with Regression Tsetlin Machine
Link: FSE Theses Repository, RUG
Engineered a novel Reinforcement Learning (RL) agent using the Regression Tsetlin Machine to achieve transparent, logic-based decision-making. Focused on replacing "black-box" neural networks with propositional logic automata to reduce computational overhead and increase model interpretability in complex environments.

PUBLICATIONS

NOISE ROBUSTNESS THROUGH ABSTRACTIONS AND ITS IMPACT ON MACHINE LEARNING | UNDER REVIEW - IEEE Transactions
Systematic analysis of data discretization techniques to enhance model robustness against noisy datasets, optimizing performance for real-world ML applications.

CACTUS: A COMPREHENSIVE ABSTRACTION AND CLASSIFICATION TOOL FOR UNCOVERING STRUCTURES | ACM Transactions on Intelligent Systems and Technology
DOI: 10.1145/3649459
Developed a high-performance analytical pipeline for healthcare data featuring optimized memory management, parallelized computation, and support for high-dimensional categorical attributes.

SANDA: A SMALL AND INCOMPLETE DATASET ANALYSER | Elsevier Journal of Information Sciences
DOI: 10.2139/ssrn.4364273
Designed an explainable analytical framework specifically for incomplete medical datasets, improving data utility in data-scarce environments.

ADAPTIVE PANDEMIC ANALYSIS: AN AGENT-BASED REINFORCEMENT LEARNING FRAMEWORK | FAIEMA 2024
DOI: Unassigned, proceedings to be published
Architected a conceptual modeling framework using Agent-Based Modeling (ABM) and RL to simulate and analyze the dynamics of complex biological systems.

PERSONAL PROJECTS

HACKEREARTH | ROAD SIGN DETECTION Computer Vision & Classification
Implemented ensemble machine learning models to classify road sign orientations, utilizing PowerBI for feature correlation analysis and data distribution mapping.

- Stack: XGBoost, Random Forests, Scikit-Learn, Numpy, PowerBI.

HACKEREARTH | AD CLICK PREDICTION Big Data & Predictive Analytics
Developed a predictive model for user ad-engagement based on historical request data and geospatial parameters.

- Stack: XGBoost, Scikit-Learn, Numpy, Jupyter.

HACKEREARTH | PRODUCT RECOGNITION Deep Learning & CNNs
Leveraged Deep Convolutional Neural Networks (Inception, VGG-19, ResNet) to automate the classification of retail inventory items.

- Stack: Keras, Scikit-Learn, Numpy, Jupyter.

HACKEREARTH | NETWORK ATTACK PREDICTION Cybersecurity AI
Built a security-focused ML model to identify server farm vulnerabilities and predict potential network attacks using anonymized traffic parameters.

- Stack: Keras, XGBoost, Scikit-Learn, Numpy.

AWESUMMLY | TOPIC MODELLING Natural Language Processing (NLP)
Constructed an NLP pipeline to automatically categorize news articles into hierarchical topic trees using advanced text embedding and classification.

- Stack: SpaCy, NLTK, GloVe, Rake, Scikit-Learn.

SKILLS

TOOLS

Analytics & Business Intelligence:
Microsoft PowerBI • Tableau • Microsoft SSRS • Matplotlib/Seaborn
Operating Systems & Environments:
Linux (Debian/Ubuntu) • Unix • Git/Version Control
Databases:
Microsoft SQL Server • MongoDB • MySQL

PROGRAMMING

Languages:
Python • C++ • C# • Julia
Frameworks (Scientific & AI):
Numpy • Scipy • Scikit-Learn • Pandas • Tensorflow • pyTsetlinMachine

CORE COMPETENCIES

Data Visualization & Automated Reporting (LaTeX/Markdown)
Multi-Agent System (MAS) Design & Implementation
Logic-Based & Symbolic AI Development
Continuous Learning & Adaptive System Analysis

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